How many of the 106 sensors can fake data give back?

SOFC plant · 39 days 2025-10-30 to 2026-03-30 · 2026-10-15

All 106, if the fake data copies real minutes (today's filler): 64 are already in its copy list, 42 need adding. **About 73, if the 50 stage sensors are made up new** and the rest must be calculated from them: 31 well, 42 roughly, 33 need their own recipe.

Way 1 - copy real minutes: 106 of 106

Way 2 - calculate from the 50 stage sensors: 73 of 106

Way 1 is like photocopying whole pages: every sensor of a real minute travels together, so they always agree, but nothing new is created. Way 2 is like guessing the rest of a page from its headline: it works for sensors that follow heat, air and gas, and fails for those that follow the operator, the gas grid or the load.

Catch: 21 of the 106 were only logged from 2026-03-17 (string power, EIRC 4 settings, FIRC 1,1,3, PIR 7,1,2). In copied data they are blank about 70 % of the time.

If you remember one thing: copying gives back all 106; calculating gives back 31 well + 42 roughly.

How we tested it

We hid one of the five run groups (F1-F5), trained one tree model per sensor on the other four groups using only the 50 stage sensors (plus stack temperature, current, heater power, flows, their 30-min change and the stage), predicted the hidden days, and compared with the real values. Repeated for all five groups.

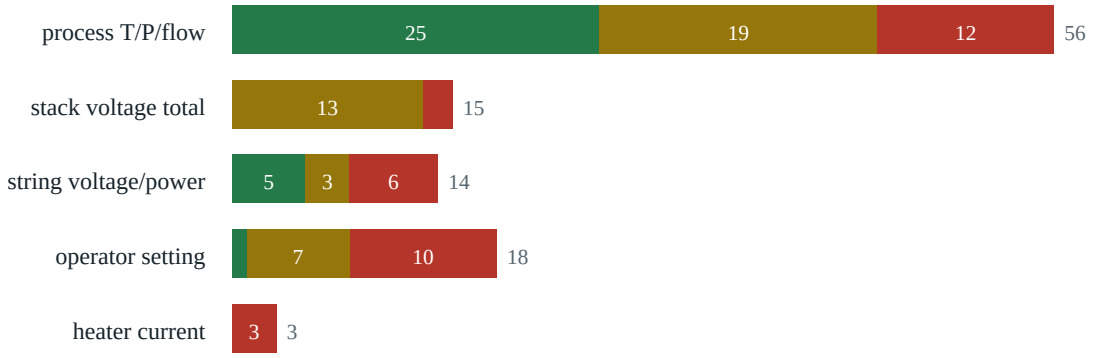
The score (R^2, share explained): 1.0 = perfect; 0 = no better than a flat line at the average; below 0 = worse.

Grade	Rule	Meaning	Sensors
A	≥ 0.90 all days and ≥ 0.70 when hot	Calculate it	31
B	≥ 0.70, or ≥ 0.90 hot only	Shape right, add real wiggle	42
C	below	Own recipe	33

Hot = stack ≥ 450 °C. Some stack voltage totals score 0.98 overall only because they are zero when cold.

If you remember one thing: the grade says how close the rebuild came on days the model never saw.

Sensor by sensor



Green A, yellow B, red C; number at the end = sensors in the family.

Sensor	Family	Grade	All days	Hot	Note
EIRC 5,1,1	heater current	C	0.64	-0.03	not in copy list
EIRC 5,1,2	heater current	C	0.63	-0.04	not in copy list
EIRC 3,1,1	heater current	C	0.60	-0.12	not in copy list
EIRC 4,2,7	operator setting	A	0.94	0.94	not in copy list; from 03-17 only
EIRC 4,2,8	operator setting	B	0.89	0.90	not in copy list; from 03-17 only
EIRC 4,4,7	operator setting	B	0.89	0.90	not in copy list; from 03-17 only
EIRC 4,5,7	operator setting	B	0.87	0.89	not in copy list; from 03-17 only
EIRC 4,3,7	operator setting	B	0.84	0.84	not in copy list; from 03-17 only
EIRC 4,2,4	operator setting	B	0.78	0.78	not in copy list; from 03-17 only
EIRC 4,4,4	operator setting	B	0.75	0.75	not in copy list; from 03-17 only
EIRC 4,1,4	operator setting	B	0.74	0.74	not in copy list; from 03-17 only
EIRC 4,3,4	operator setting	C	0.51	0.48	not in copy list; from 03-17 only
EIRC 4,1,7	operator setting	C	-0.02	-0.02	not in copy list; from 03-17 only
EIRC 4,3,5	operator setting	C	-0.06	-0.01	not in copy list
EIRC 4,2,5	operator setting	C	-0.94	-0.63	not in copy list
EIRC 4,1,5	operator setting	C	-1.31	-0.79	not in copy list
EIRC 4,3,8	operator setting	C	-4.44	-5.26	not in copy list; from 03-17 only
EIRC 4,5,4	operator setting	C	-4.71	-5.36	not in copy list; from 03-17 only
EIRC 4,4,8	operator setting	C	-5.26	-6.53	not in copy list; from 03-17 only
EIRC 4,5,8	operator setting	C	-5.34	-6.62	not in copy list; from 03-17 only
EIRC 4,5,5	operator setting	C	-12.57	-11.63	not in copy list
TIRC 2,06	process T/P/flow	A	1.00	0.98	
TIR 5,4,2	process T/P/flow	A	1.00	0.99	
PIR 5,3,1	process T/P/flow	A	1.00	0.99	
PIR 5,3,2	process T/P/flow	A	1.00	0.99	
PIR 1,1,2	process T/P/flow	A	0.99	0.99	
TIRZ 8,1,1	process T/P/flow	A	0.99	0.85	
PIR 2,01	process T/P/flow	A	0.99	0.99	
TIR 2,05	process T/P/flow	A	0.99	0.95	

Sensor	Family	Grade	All days	Hot	Note
PIR 7,1,1	process T/P/flow	A	0.99	0.98	
TIRC 5,2,1	process T/P/flow	A	0.99	0.91	
PIR 1,2,2	process T/P/flow	A	0.99	0.90	
PIR 7,1,2	process T/P/flow	A	0.99	0.98	from 03-17 only
TIRC 5,10	process T/P/flow	A	0.98	0.93	
TIRC 5,3,1	process T/P/flow	B	0.97	0.68	
TIRZ 8,2,2	process T/P/flow	B	0.96	0.60	
FIRC 1,1,3	process T/P/flow	A	0.96	0.89	from 03-17 only
TIRC 5,11	process T/P/flow	A	0.96	0.90	
TIRC 3,10	process T/P/flow	A	0.96	0.95	
PIR 6,01	process T/P/flow	A	0.95	0.97	
TIRC 3,11	process T/P/flow	A	0.95	0.91	
TIRZ 8,2,5	process T/P/flow	B	0.94	0.31	
PIR 1,3,1	process T/P/flow	A	0.94	0.80	
TIRC 3,07	process T/P/flow	A	0.94	0.95	
PIR 3,01	process T/P/flow	A	0.94	0.94	
FIRC 1,4	process T/P/flow	A	0.94	0.94	
PIR 1,4,2	process T/P/flow	A	0.94	0.94	
TIRC 5,5,2	process T/P/flow	B	0.93	0.66	
PIR 7,02	process T/P/flow	A	0.93	0.96	
TIR 2,04	process T/P/flow	B	0.91	0.68	
TIRC 5,5,1	process T/P/flow	A	0.91	0.76	
PIR 7,03	process T/P/flow	B	0.90	0.93	
TIR 3,06	process T/P/flow	B	0.87	0.95	
TIRZ 8,1,10	process T/P/flow	B	0.87	0.42	
TIR 7,04	process T/P/flow	B	0.83	0.45	
TIRC 3,03	process T/P/flow	B	0.78	0.87	
TIR 7,3,2	process T/P/flow	B	0.78	0.61	
TIR 7,3,1	process T/P/flow	B	0.78	0.61	
TIR 6,01	process T/P/flow	B	0.77	0.90	
PIR 1,3,2	process T/P/flow	B	0.75	0.41	
TIR 3,01	process T/P/flow	B	0.74	0.66	
TIR 3,05	process T/P/flow	B	0.73	0.82	
TIR 1,2,2	process T/P/flow	B	0.71	0.96	
TIR 3,04	process T/P/flow	B	0.70	0.97	
TIRC 6,1,1	process T/P/flow	C	0.66	0.57	
TIRZ 8,2,4	process T/P/flow	C	0.59	0.57	
PIR 1,5	process T/P/flow	C	0.48	0.61	
TIR 3,02	process T/P/flow	C	0.12	-0.57	
PIRZ 8,1,2	process T/P/flow	C	0.07	-0.30	
QIR 1,4	process T/P/flow	C	-0.01	-2.18	not in copy list

Sensor	Family	Grade	All days	Hot	Note
PIR 1,1,3	process T/P/flow	C	-0.17	-0.08	
TIR 4,1	process T/P/flow	C	-0.18	-0.03	
PIR 1,4,1	process T/P/flow	C	-0.24	0.00	
PIR 1,1,1	process T/P/flow	C	-0.34	-0.01	
PIR 1,2,1	process T/P/flow	C	-0.39	-0.08	
PIR 3,02	process T/P/flow	B	-1.19	0.93	
TIR 2,1,1	process T/P/flow	C	-3.76	-0.74	
EIR 4,5,15	stack voltage total	B	0.99	0.51	not in copy list
EIR 4,1,18	stack voltage total	B	0.99	0.22	not in copy list
EIR 4,2,15	stack voltage total	B	0.99	0.46	not in copy list
EIR 4,3,18	stack voltage total	B	0.99	0.25	not in copy list
EIR 4,5,12	stack voltage total	B	0.99	0.47	not in copy list
EIR 4,4,15	stack voltage total	B	0.99	0.49	not in copy list
EIR 4,3,12	stack voltage total	B	0.99	-0.05	not in copy list
EIR 4,2,12	stack voltage total	B	0.99	-0.08	not in copy list
EIR 4,3,15	stack voltage total	B	0.98	-0.11	not in copy list
EIR 4,4,12	stack voltage total	B	0.98	0.18	not in copy list
EIR 4,2,18	stack voltage total	B	0.96	-0.32	not in copy list
EIR 4,5,18	stack voltage total	B	0.94	-0.03	not in copy list
EIR 4,4,18	stack voltage total	B	0.86	-0.32	not in copy list
EIR 4,1,15	stack voltage total	C	0.56	-0.65	not in copy list
EIR 4,1,12	stack voltage total	C	0.56	-0.76	not in copy list
EIRC 4,3,3	string voltage/power	A	0.93	0.93	not in copy list; from 03-17 only
EIRC 4,1,3	string voltage/power	A	0.92	0.92	not in copy list; from 03-17 only
EIRC 4,4,3	string voltage/power	A	0.92	0.92	not in copy list; from 03-17 only
EIRC 4,2,3	string voltage/power	A	0.91	0.91	not in copy list; from 03-17 only
EIRC 4,5,3	string voltage/power	A	0.90	0.90	not in copy list; from 03-17 only
EIRC 4,1,1	string voltage/power	B	0.49	0.95	
EIRC 4,2,1	string voltage/power	B	0.12	0.96	
EIRC 4,4,1	string voltage/power	B	0.04	0.96	
EIRC 4,1,9	string voltage/power	C	-0.15	-0.78	
EIRC 4,2,9	string voltage/power	C	-0.15	-0.81	
EIRC 4,3,9	string voltage/power	C	-0.19	-0.89	
EIRC 4,4,9	string voltage/power	C	-0.23	-0.96	
EIRC 4,5,9	string voltage/power	C	-0.27	-0.32	
EIRC 4,3,1	string voltage/power	C	-0.34	0.89	

If you remember one thing: heat, air and gas sensors come back; operator- and load-driven ones need a recipe.

What to do with the 33 hard ones

Group	n	Recipe	Sensors
Operator settings	10	Write from the stage plan: copy the real setting sequence	EIRC 4,1,7, EIRC 4,5,5, EIRC 4,3,5, EIRC 4,1,5, EIRC 4,2,5, EIRC 4,3,8, EIRC 4,3,4, EIRC 4,5,8, EIRC 4,4,8, EIRC 4,5,4
Cell minimum / string voltage	6	Small voltage-current line per string from the ~950 load minutes	EIRC 4,3,1, EIRC 4,1,9, EIRC 4,2,9, EIRC 4,4,9, EIRC 4,3,9, EIRC 4,5,9
Supply pressures	7	Slow random drift around the real level (follow gas grid)	PIR 1,1,1, PIR 1,1,3, PIR 1,2,1, PIR 1,4,1, PIR 1,5, QIR 1,4, PIRZ 8,1,2
Other temperatures	5	Keep copying from the real episode	TIR 2,1,1, TIR 4,1, TIRZ 8,2,4, TIR 3,02, TIRC 6,1,1
Heater currents	3	Copy, or link to that heater's own on/off	EIRC 5,1,2, EIRC 5,1,1, EIRC 3,1,1
Stack 1 voltage totals	2	Copy from the real episode	EIR 4,1,12, EIR 4,1,15

Suggested order

Step	Action
Now	Add the 42 missing sensors to the filler's copy list -> all 106 in fake data
Next	When curves are stretched or new: calculate the 73 A and B sensors, add real wiggle to B
Then	Give the C sensors their recipe
Check	Run the 50-sensor stage checker; flag blocks under 80 %

If you remember one thing: adding the 42 missing sensors to the copy list alone gives all 106 today.